# Jinx: Catching Duplicate Pull Requests from Coding Agents

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## Abstract

Coding agents open many pull requests (PRs), and some repeat work that is already open in the same repository. Jinx is a GitHub Action and command-line tool that compares a new PR with the repository's open PRs and posts one comment listing likely duplicates, each with plain-language reasons (shared files, overlapping lines, a shared issue, similar text after removing agent template text). It runs on a CPU in 22 ms per PR (median), needs no API key, and sends no code to an external service. On 342 cross-author duplicate pairs from real agent PRs, scored against each repository's open queue, Jinx ranks the true duplicate first for 0.901 of pairs, against 0.778 for the best existing open-source duplicate-PR bot and 0.810 for a classic INTRUDE-style detector, with 2.5% false alarms. Video: [TODO: URL]. Tool: [TODO: URL].

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## 1 Introduction

Coding agents such as Codex, Copilot, Devin, Cursor, and Claude Code now open hundreds of thousands of PRs [4]. Duplicate or superseded work is a leading reason why agent PRs are closed without merging: 142 of 600 failed agent PRs in one study [2], and "resolved by another PR" tops the list in another [1]. Duplicates happen when several agents are assigned the same issue, or when one agent replaces its own PR instead of updating it. Each duplicate wastes a maintainer's review.

GitHub only lets a maintainer mark a duplicate by hand. Commercial AI review tools document duplicate detection for issues, not PRs; CodeRabbit's "Related PRs" lists PRs that touched the same code as context, not as a duplicate alarm. Small open-source bots (contrib-guard, Simili, PRSense, PRism, vector-triage, doppelgänger[1]) compare PR titles and descriptions with text or embedding similarity, and some need an LLM API key. Research detectors [3, 5–8] were built for human PRs and are not packaged as maintained tools.

Jinx fills this gap: it uses the code changes as well as the text, explains each match, runs inside the repository's own CI, and is tuned and evaluated on agent PRs.

## 2 Jinx

Fig. 1 shows the pipeline. **Input.** In the Action, Jinx reads the new PR and up to 200 open PRs (titles, descriptions, linked issues, changed files and patches) through the GitHub API with the workflow's own token. The CLI reads the same fields from JSON files. **Template stripping.** Agents repeat boilerplate (checklists, "Generated with …" footers). Jinx learns lines that recur across a repository's PRs and removes them before comparing text. **Signals.** For each open PR Jinx computes TF-IDF similarity of title and body, an optional local MiniLM embedding similarity, overlap of changed files, overlap of changed line ranges, share of identical added lines, shared linked issues, explicit references between the PRs, same author or agent, time gap, and whether the candidate is still open. **Score and reasons.** A logistic model, trained on our benchmark and shipped with the package, combines the signals. The score threshold of 0.5 is set to the median top-1 score of cross-author duplicates. Each signal that contributes is turned into a sentence. **Output.** The CLI prints a ranked list; the Action posts or updates a single comment and never closes or labels PRs.

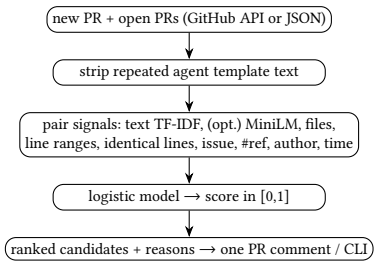



**Figure 1: The Jinx pipeline.**

```
$ jinx check new.json --open-queue open.jsonl
PR #44134: checked 19 open PRs in 12 ms.
  DUPLICATE? #44110  score 1.00  refactor(tvp): use
             capture_outputs for output tracing
    - one PR links to the other (#44110)
    - shares 1 of 1 changed files (.../modeling_tvp.py)
    - 100% of changed line ranges overlap
    - 100% of added lines are identical
    - similar title/description after removing
      agent template text (0.92)
```

**Figure 2: Jinx on a real agent PR from huggingface/transformers (AIDev).**

**Usage.** Install with `pip install jinx-pr`, or add the Action to a workflow:

```
on:
  pull_request_target:
    types: [opened, reopened]
permissions:
  contents: read
  pull-requests: write
  issues: write
jobs:
  jinx:
```

[1] https://github.com/dannyl1u/doppelganger

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**Table 1: Cross-author pairs ($n$ = 342); false alarms on 2965 PRs with no duplicate.**

| Strategy | R@1 | R@5 | MRR | False alarms % [95% CI] |
|---|---|---|---|---|
| Jinx (ours) | 0.901 | 0.971 | 0.932 | 2.5 [1.9, 3.3] |
| All signals, LR | 0.906 | 0.974 | 0.935 | 1.6 [1.1, 2.1] |
| Jinx + BGE rerank + Qwen-7B | 0.901 | 0.974 | 0.931 | 2.8 [2.2, 3.5] |
| Claude Haiku 4.5 + Jinx | 0.898 | 0.971 | 0.932 | 1.8 [1.3, 2.3] |
| Qwen2.5-14B + Jinx | 0.871 | 0.971 | 0.917 | 2.9 [2.3, 3.7] |
| Claude Haiku 4.5 alone | 0.842 | 0.971 | 0.902 | 5.4 [4.6, 6.1] |
| INTRUDE + time [7, 8] | 0.810 | 0.953 | 0.873 | 4.6 [3.7, 5.7] |
| contrib-guard (bot) | 0.778 | 0.918 | 0.843 | 11.7 [8.9, 14.6] |
| Links only | 0.839 | 0.924 | 0.882 | 3.8 [3.0, 4.8] |
| TF-IDF [5] | 0.757 | 0.921 | 0.832 | 9.4 [7.8, 10.9] |
| BGE-small embedding | 0.763 | 0.915 | 0.830 | 6.4 [5.3, 7.5] |

```
runs-on: ubuntu-latest
steps:
  - uses: ABCR-Labs/jinx-pr@v1
```

[TODO: confirm final repository name and tag]

## 3 Evaluation

We built a live-queue benchmark from AIDev [4]: for each of 1867 PR pairs that a comment marks as duplicate or superseded, the new PR is compared against the PRs open in its repository at that moment, and 2965 PRs with no duplicate measure false alarms. Table 1 compares Jinx with the existing bots, a classic detector, embeddings, and LLM judges on the hardest case, duplicates opened by different authors. Jinx beats the best bot on Recall@5 by +5.3 points [+1.2, +9.7] and on Recall@1 by +12.3 points, and beats INTRUDE + time on Recall@1 by +9.1 points [+5.0, +13.1] with fewer false alarms. LLM judges (Claude Haiku 4.5, Qwen2.5) on top of Jinx add cost and latency but no accuracy. Jinx needs 22 ms per PR (p90 229 ms) on a CPU, with no GPU. A companion empirical study reports all 66 strategies [TODO: cite once public].

## 4 Limitations

Jinx misses many duplicates that touch different files (cross-author Recall@1 0.655), and its Recall@5 is not clearly better than the classic detector's. Our labels come from PR comments, and an LLM audit judged 84.8% of cross-author labels to be true duplicates or replacements. We have not yet measured Jinx on live repositories; [TODO: sandbox and pilot deployment results].

## 5 Availability

Jinx is open source (MIT): [TODO: repository URL], PyPI `jinx-pr`, GitHub Marketplace [TODO: listing]. The benchmark, scripts, and results are archived at [TODO: DOI]. Demo video: [TODO: URL].

## References


[1] Alam, Mondal, and Chanchal K. Roy. 2026. Why Are AI Agent Involved Pull Requests (Fix-Related) Remain Unmerged? arXiv:2602.00164

[2] R. Ehsani et al. 2026. Where Do AI Coding Agents Fail? An Empirical Study of Failed Agentic Pull Requests in GitHub. arXiv:2601.15195 MSR 2026.

[3] Jing Jiang, Qiudi Li, Jin Guo, Zhenrui Li, Li Zhang, and Ge Li. 2023. DupHunter: Detecting Duplicate Pull Requests in Fork-Based Development. *IEEE Trans. Softw. Eng.* (2023). doi:10.1109/TSE.2023.3235942

[4] Hao Li et al. 2026. AIDev: Studying AI Coding Agents on GitHub. arXiv:2602.09185

[5] Zhixing Li, Gang Yin, Yue Yu, Tao Wang, and Huaimin Wang. 2017. Detecting Duplicate Pull-Requests in GitHub. In *Proc. Internetware.* doi:10.1145/3131704.3131725

[6] Zhifang Liao, Yuanyuan Liu, Yuqi Lan, and Yan Sun. 2024. DupLLM: Duplicate Pull Requests Detection Based on Large Language Model. In *Proc. APSEC.* doi:10.1109/APSEC65559.2024.00023

[7] Luyao Ren, Shurui Zhou, Christian Kästner, and Andrzej Wąsowski. 2019. Identifying Redundancies in Fork-based Development. In *Proc. SANER.* doi:10.1109/SANER.2019.8668023

[8] Qingye Wang, Bowen Xu, Xin Xia, Ting Wang, and Shanping Li. 2019. Duplicate Pull Request Detection: When Time Matters. In *Proc. Internetware.* doi:10.1145/3361242.3361254